

DATABASE SYSTEMS

Week 1,2

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Switch off mobile phones during lectures, or put them into silent mode

AUTHOR SLIDES

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Database Systems

Chapter 1

Introduction to Databases

Agenda

- File-based Systems
- History of Database
- **Database Management Systems (DBMS)**

File-based Definition

- Program defines and manages it's own data

Limitations of File-based

- Separation and isolation
- Duplication
- Program & data dependence
- Fixed queries
- Lot of application programs to manage

Database

- Definition
 - A collection of self-describing and integrated data files
- System catalog
 - Meta data
 - Data dictionary
 - Overhead data
- Data abstraction

Database Management System Facility

- Data definition language (DDL)
- Data manipulation language (DML)
- Structured query language (SQL)
- Security system
- Integrity system
- Concurrency control system
- Backup & recovery system
- View mechanism

DBMS Environment

- Hardware
 - Client-server architecture
- Software
 - dbms, os, network, application
- Data
 - Schema, subschema, table, attribute
- People
 - Data administrator & database administrator
 - Database designer: logical & physical
 - Application programmer
 - End-user: naive & sophisticated
- Procedure
 - Start, stop, log on, log off, back up, recovery

Advantages of DBMS

- Control redundancy
- Consistency
- Integrity
- Security
- Concurrency control
- Backup & recovery
- Data standard
- More information
- Data sharing & conflict control
- Productivity & accessibility
- Economy of scale
- Maintenance

Limitations of DBMS

- Complexity
- Size
- Cost
 - Software
 - Hardware
 - Conversion
- Performance
- Vulnerability

Points to Remember

- File-based Systems
- History of Database
- **Database Management Systems (DBMS)**

Chapter 2

Database Environment

Agenda

- Three-Level ANSI-SPARC Architecture
- Database Languages
- Data Models
- Functions of DBMS
- Components of DBMS
- Teleprocessing

Three-Level ANSI-SPARC Architecture

- External level
 - User's view
 - Data gathering
- Conceptual level
 - Organization view (entity, attribute, & relationship)
 - Constraints and security
 - Entity-relationship diagram
 - Normalization
- Internal level
 - Physical presentation
 - Storage, index, compression, & encryption
 - File organization

Database Schemas

- Database schema (intension) & database instance (extension)
 - External schemas (subschemas)
 - Conceptual schema (database schema)
 - Internal schema

Mapping

- External/conceptual mapping
- Conceptual/internal mapping
- Example

Data Independence

- Logical data independence
- Physical data independence

Database Languages

- Data Definition Language (DDL)
 - Structured Query Language (SQL)
- Data Manipulation Language (DML)
 - Procedural DML
 - Non-procedural DML
 - Structured Query Language (SQL)
 - Query-by-Example (QBE)
- Fourth-Generation Language
 - Form, report, graphics, & application generators

Data Models

- Definition
 - Integrated concept for describing data, relationships and constraints
- Types
 - Object-based data models
 - Record-based data models
 - Physical data models (internal structure, ordering, & paths)

Object-Based Data Models

- Entity-relationship
 - Entity, attribute, relationship
 - Usage: documentation
- Object-oriented
 - Object, class, subclass, inheritance, state (attributes), behavior (methods or actions), encapsulation, message, polymorphism
 - Usage: building software

Record-Based Data Models

- Relational data model
- Network model
- Hierarchical Model

Relational Data Model

- Terminology
 - Relations
 - Attributes
 - Tuples
- Record relationship
 - One-to-many relationship

Network Model

- CODASYL DBTG
- Terminology
 - Data item & group item
 - Record type: owner, member
 - Set type: optional, mandatory, permanent
- Record relationship
 - Many-to-many relationship
 - Link or pointer between set owner and set member
- Usage
 - Large volume transaction processing

Hierarchical Model

- IBM
- Information Management Systems
- Terminology
 - Tree, general tree, & subtree
 - Nodes, root node, parent node, and child node
 - Segment type (root, parent, child)
- Record relationship
 - One-to-many relationship
 - Link or pointer between parent node and child node
 - No many-to-many relationship
- Usage
 - Large volume transaction processing

Functions of a DBMS

- Data storage, retrieval and update
- A user-accessible catalog
- Transaction support
- Concurrency control
- Recovery
- Authorization
- Integrity
- Data independence
- Data communication
- Other Utilities

Components of a DBMS

- Query processor
- Database manager
- File manager
- DML preprocessor
- DDL compiler
- Catalog manager

Components of Database Manager

- Authorization control
- Command processor
- Integrity checker
- Query optimizer
- Transaction manager
- Scheduler
- Recovery manager
- Buffer manager

Two-tier client-server architecture

- Client: user interface, business and data processing logic
- Database server: data validation and database access
- Advantage
 - Accessing to distributed database
 - Increasing performance and consistency
 - Lowering server and communication cost
 - Supporting open systems architecture

Two-tier client-server architecture

- Three-tier client-server architecture
 - Client: user interface
 - Application server: business and processing logic
 - Database server: data validation and database access
 - Advantage
 - Reducing client cost
 - Software distribution
 - Maintenance cost
 - Balancing load

Two-tier client-server architecture

- Transaction processing (TP) monitor
 - Transaction manager between client and server
 - Advantage
 - Transaction routing
 - Distributed transaction for load balancing

Points to Remember

- Three-Level ANSI-SPARC Architecture
- Database Languages
- Data Models
- Functions of DBMS
- Components of DBMS
- Teleprocessing

Thanks